

Bay - Test 1 field report

Melbourne CBD field test | Short tracking record | Updated 24 September 2026

Purpose

Record the supplied screenshots and photos, preserve the evidence trail, and separate verified fixes from mitigations that still need another field test. A photograph proves a mismatch at that location; it does not establish citywide accuracy.

Current accuracy picture

Area	Evidence / current level
Sensor identity	4,986 of 5,794 live rows had a heartbeat inside Bay's 20-minute display window at the check (86.1%). Exact-ID matching is the trusted path; coordinate fallback is unverified.
Occupancy truth	Not measurable as a percentage from screenshots alone. The app now expires stale signals, marks conflicts amber and distinguishes vacant from legally parkable.
Rule truth	Low for segment-derived mixed-sign blocks. The imported sign/zone datasets contain no sign coordinates or arrows, so the app now labels those rules as needing verification.
Map coverage	The old 3,000-row cap is removed; offset paging and refresh are implemented.
Orientation	Manual rotation, north reset, touch rotation and permission-aware compass following are implemented; device testing remains.

Issue register

1. Pins said vacant while cars were visible	Observed at Spencer / La Trobe / Queen Street. Sensors report bay presence periodically; they are not a camera.	Mitigated. Exact-ID provenance is retained; stale, approximate and contradictory signals become uncertain/unknown. Repeat field test required.
2. Four cars versus three occupied pins	A screenshot cannot establish whether a sensor changed just before the photo.	Mitigated. One-minute refresh, expiry and conflict display added.
3. Whole rows disagreed with the street	Sensor coordinates and legal bay geometry are separate sources.	Mitigated. Only exact council ID is trusted for sensor identity; coordinate matches are explicitly unverified.
4. 2P meter shown as loading zone	The Queen / A'Beckett photo shows a 2P meter sign with a no-parking arrow. Melbourne zone/segment exports have no sign coordinates or arrows.	Safely contained, not fully solved. Segment rules show "Rules need verification" and cannot present confident legality.
5. Turning / unclear bay treated as a spot	The bay export does not encode all legal stopping geometry or turning-area semantics.	Open. Requires a bay-eligibility audit; occupancy is not proof of legality.
6. Map did not rotate	Leaflet does not provide compass following by default.	Fixed in source. Plugin, manual rotation, north reset, touch rotation and permission-aware compass mode added. Device testing remains.
7. "3000 spots found" hid results	The client used a 3,000-row cap.	Fixed in source. Offset paging, deduplication, one-minute refresh and update time added.

Why Melbourne has sensors

Council material describes in-ground sensors as tools for parking-time enforcement, identifying overstays, improving patrol efficiency, monitoring occupancy and turnover, checking service reliability, and informing transport policy. The public feed exposes presence/absence, location and kerbside ID. Bay can use that feed as a secondary signal, but it cannot replace the sign or prove legal permission.

Can Bay work manually across Australia?

Yes: nearby users can report occupied/vacant with GPS proximity, freshness limits and corroboration. It scales without council hardware, but accuracy depends on local participation. The safe product rule is "unknown" when reports are absent or conflicting, never "vacant" by default. Melbourne sensors remain optional and provenance-labelled.

Dataset completeness

Council snapshots remain under datasets/. A manifest with source URLs and SHA-256 hashes and a non-overwriting archive helper were added. Newly checked sensor and sign exports are archived under datasets/research-2026-09-24/. The sign-plate datasets contain zone/rule fields but no sign coordinates; this is the key limitation behind issue 4.

Next verification pass

Repeat the same streets after the new build. For each photographed bay, record GPS, sign arrow, council bay reference, app legal status, app occupancy status, sensor age and outcome. Calculate separate percentages for sensor identity, occupancy agreement and legal-rule agreement; do not combine them.